

Study of different types of network cables

a) Understand different types of network cables

Different type of cables used in networking are

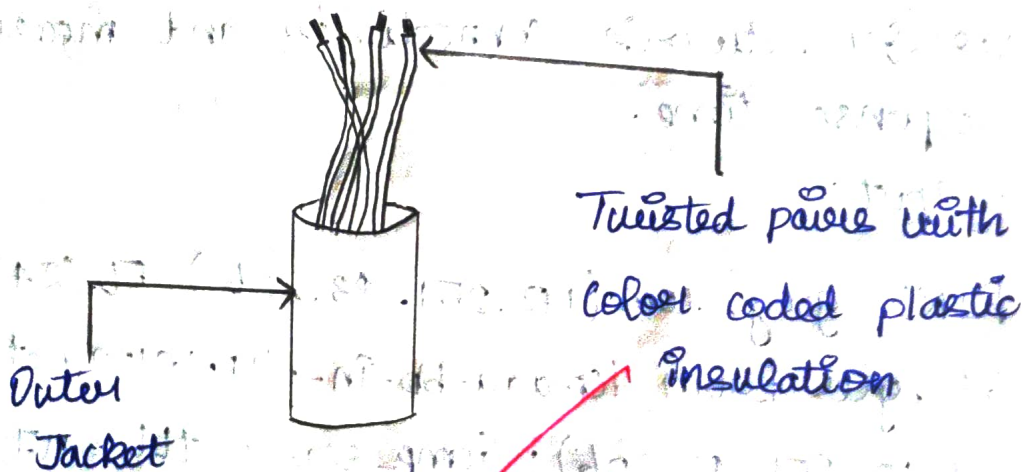
1. Unshielded Twisted Pair (UTP) cable
2. Shielded Twisted Pair (STP) cable
3. Coaxial Cable
4. Fibre Optic Cable

1. UTP Cable

Category 3 - 10bps

Category 5 - Upto 100 Mbps

Category 5e - 1Gbps

Advantages:

- *) Cheaper in cost
- *) Easy to install as they have a smaller overall diameter.

Disadvantages:-

*) More prone to (EMI) Electromagnetic Interference and noise.

Applications:-

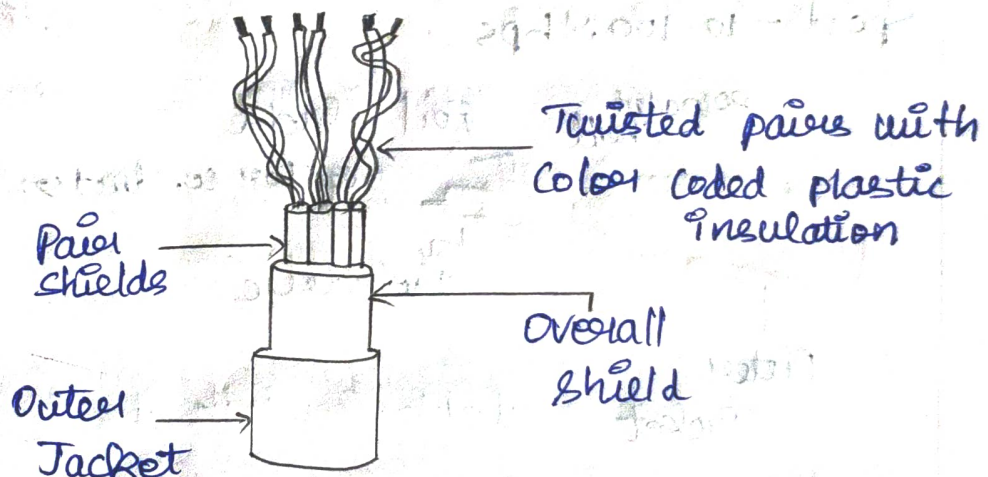
*) 10Base-T Ethernet

*) Fast Ethernet, Gigabit Ethernet

~~*) Fast~~

2. STP :-

Category 6, 6a - 10Gbps



SSTP

Category 7 - 10Gbps

Advantages:-

*) Shielded

*) Faster than UTP

*) Less susceptible to noise and

Interference

Disadvantages:-

*) Expensive

*) Greater installation effort

Applications

* Gigabit Ethernet, 10G Ethernet (55m)
widely used in data centers.

* Gigabit Ethernet, 10G Ethernet (100m)

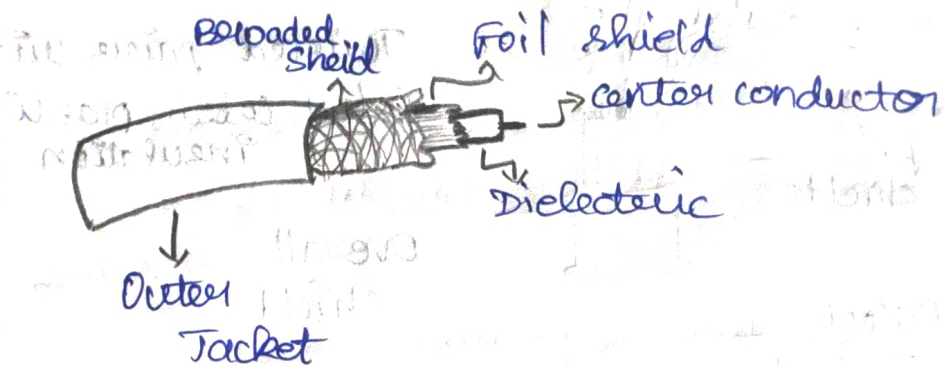
3. Coaxial Cable

RG-6

RG-59

RG-11

Speed - 10-100 Mbps.



Advantages:-

- * High bandwidth
- * Immune to interference
- * Low Loss bandwidth
- * Versatile.

Disadvantages:-

- * Limited distance
- * Cost
- * Size is bulky.

Applications:

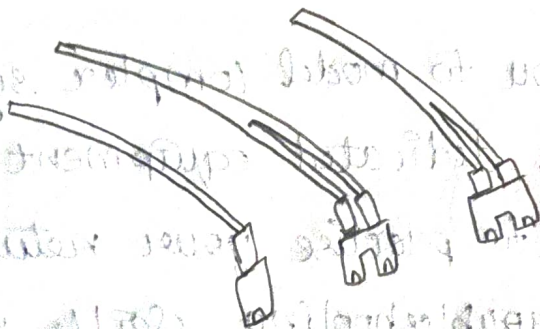
- * Speed of signal is 500m
- * Television network
- * High speed internet connections

4. Fibre Optic Cable

Single mode

Multi mode

- 100Gbps



Advantages:

- * High speed
- * High bandwidth
- * High security
- * Long distance

Disadvantages:

- * Expensive
- * Requires skilled installers

Application

- * Maximum distance of fibre optics

Cable is around 100 metres.

Result:-

Hence studied all the types of cables.